

# TFM - Stacked $y$ -maps and $\mu y$ -maps

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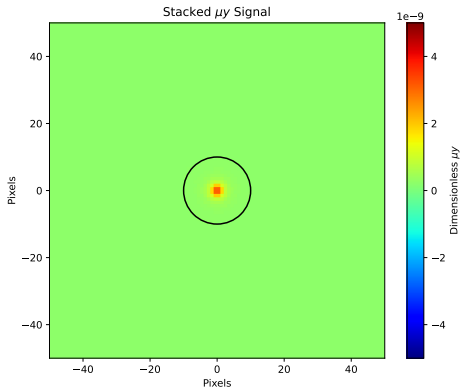
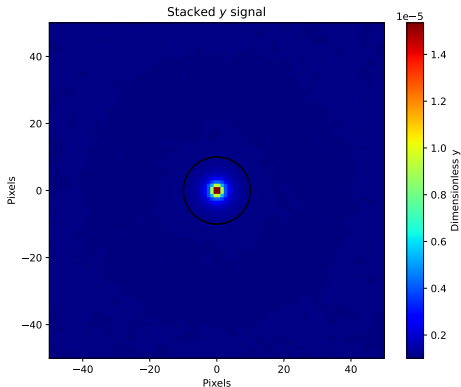
# Input stacked zoomed maps

4 layers: tSZ,  $\mu$ -distortion, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ )

$\mu = 2 \times 10^{-4}$ , freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$  pixels, stacked maps =  $10^4$

Input Data stacked patch of 10000 brightest clusters,  $\mu = 2\text{e-}04$ , base sensitivity =  $0.20 \mu K \cdot \text{arcmin}$ ,  $r = 10$



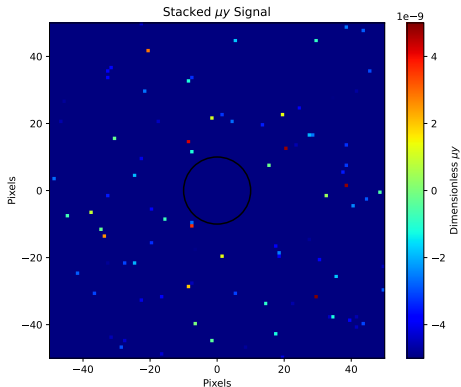
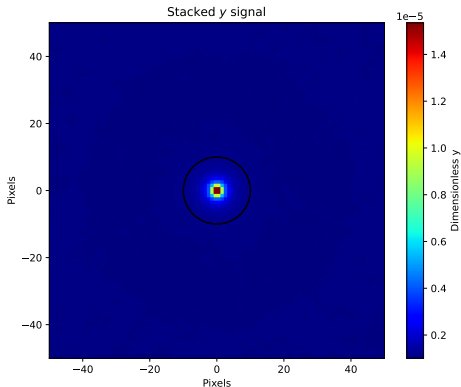
# ILC stacked zoomed maps

4 layers: tSZ,  $\mu$ -distortion, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ )

$\mu = 2 \times 10^{-4}$ , freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$  pixels, stacked maps =  $10^4$

P-ILC stacked patch of 10000 brightest clusters,  $\mu = 2\text{e-}04$ , base sensitivity =  $0.20 \mu K \cdot \text{arcmin}$ ,  $r = 10\text{p}$



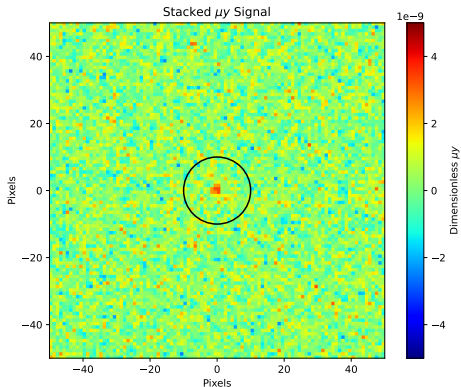
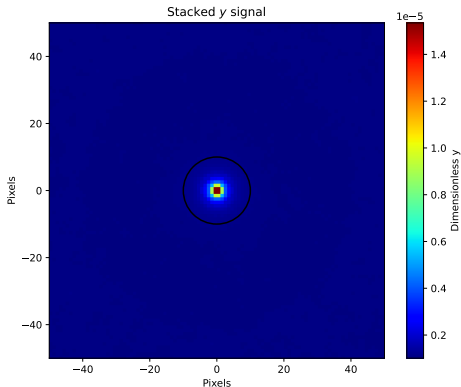
# CILC stacked zoomed maps

4 layers: tSZ,  $\mu$ -distortion, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ )

$\mu = 2 \times 10^{-4}$ , freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$  pixels, stacked maps =  $10^4$

P-CILC stacked patch of 10000 brightest clusters,  $\mu = 2\text{e-}04$ , base sensitivity =  $0.20 \mu K \cdot \text{arcmin}$ ,  $r = 10\text{p}$



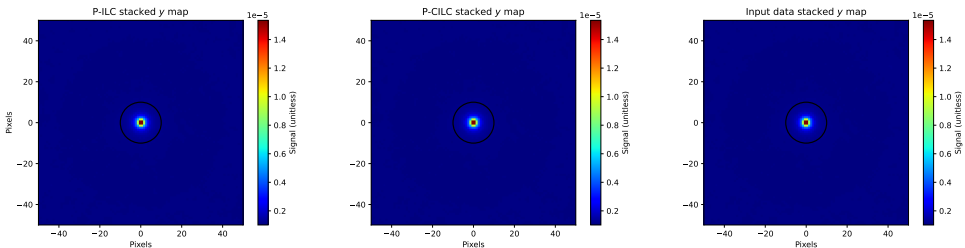
# Comparison $y$ stacked zoomed maps

4 layers: tSZ,  $\mu$ -distortion, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ )

$\mu = 2 \times 10^{-4}$ , freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$  pixels, stacked maps =  $10^4$

Comparison of  $y$  map stacked patches of  $1e+04$  brightest clusters,  $\mu$  value =  $2e-04$ , base sensitivity =  $0.20 \mu K \cdot \text{arcmin}$ ,  $r = 10p$



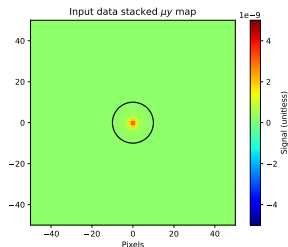
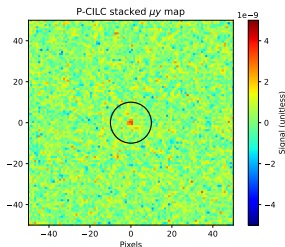
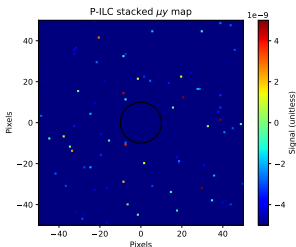
# Comparison $\mu\gamma$ stacked zoomed maps

4 layers: tSZ,  $\mu$ -distortion, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ )

$\mu = 2 \times 10^{-4}$ , freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$  pixels, stacked maps =  $10^4$

Comparison of  $\mu\gamma$  map stacked patches of  $1e+04$  brightest clusters,  $\mu$  value =  $2e-04$ , base sensitivity =  $0.20 \mu K \cdot \text{arcmin}$ ,  $r = 10p$



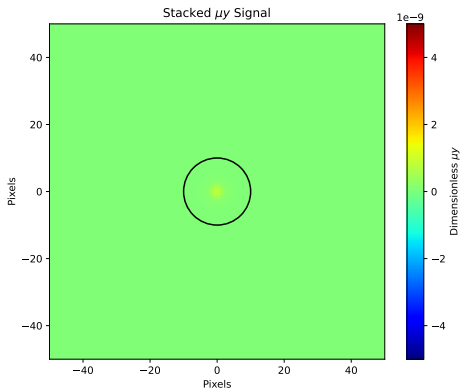
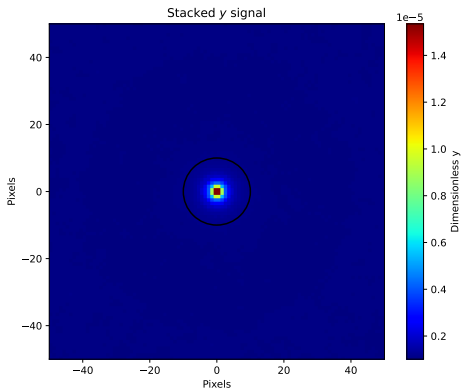
# Input stacked zoomed maps

4 layers: tSZ,  $\mu$ -distortion, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ )

$\mu = 5 \times 10^{-5}$ , freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$  pixels, stacked maps =  $10^4$

Input Data stacked patch of 10000 brightest clusters,  $\mu = 5\text{e-}05$ , base sensitivity =  $0.20 \mu K \cdot \text{arcmin}$ ,  $r = 10$



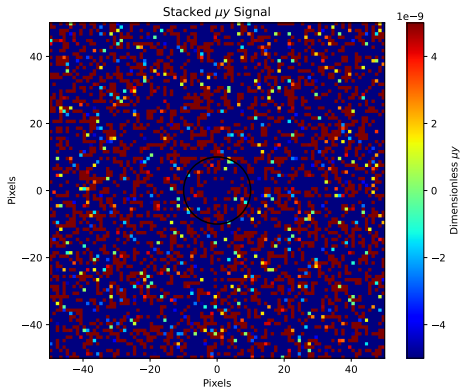
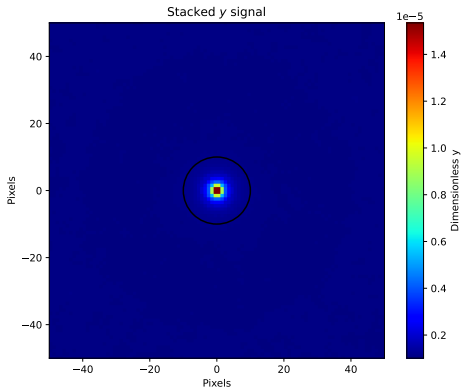
# ILC stacked zoomed maps

4 layers: tSZ,  $\mu$ -distortion, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ )

$\mu = 5 \times 10^{-5}$ , freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$  pixels, stacked maps =  $10^4$

P-ILC stacked patch of 10000 brightest clusters,  $\mu = 5\text{e-}05$ , base sensitivity =  $0.20 \mu K \cdot \text{arcmin}$ ,  $r = 10\text{p}$



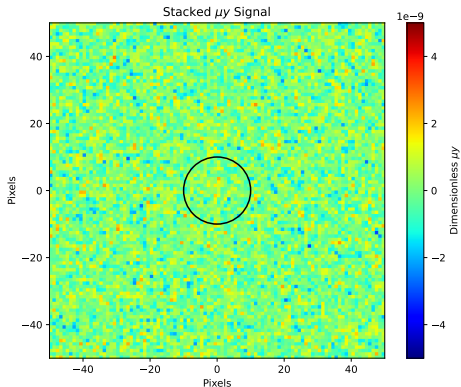
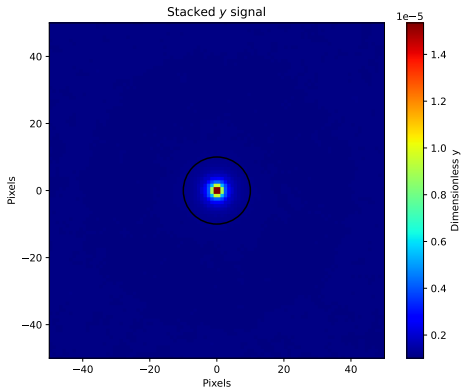
# CILC stacked zoomed maps

4 layers: tSZ,  $\mu$ -distortion, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ )

$\mu = 5 \times 10^{-5}$ , freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$  pixels, stacked maps =  $10^4$

P-CILC stacked patch of 10000 brightest clusters,  $\mu = 5\text{e-}05$ , base sensitivity =  $0.20 \mu K \cdot \text{arcmin}$ ,  $r = 10\text{p}$



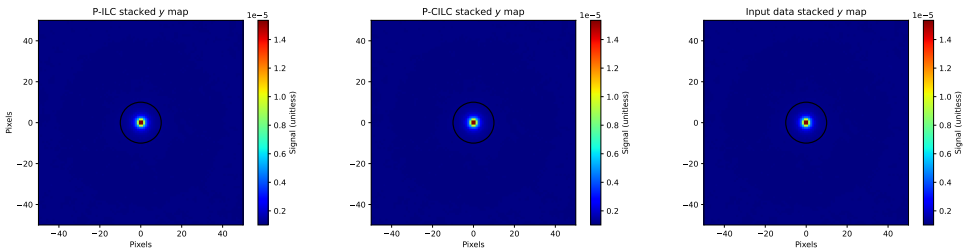
# Comparison $y$ stacked zoomed maps

4 layers: tSZ,  $\mu$ -distortion, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ )

$\mu = 5 \times 10^{-5}$ , freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$  pixels, stacked maps =  $10^4$

Comparison of  $y$  map stacked patches of  $1e+04$  brightest clusters,  $\mu$  value =  $5e-05$ , base sensitivity =  $0.20 \mu K \cdot \text{arcmin}$ ,  $r = 10p$



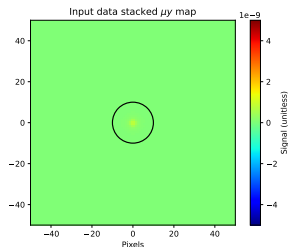
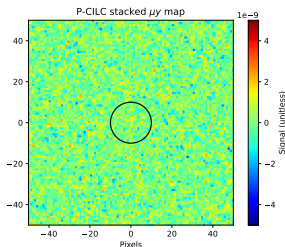
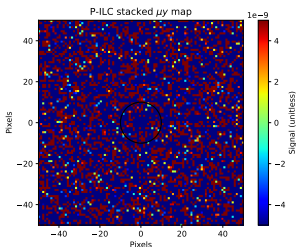
# Comparison $\mu\gamma$ stacked zoomed maps

4 layers: tSZ,  $\mu$ -distortion, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ )

$\mu = 5 \times 10^{-5}$ , freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$  pixels, stacked maps =  $10^4$

Comparison of  $\mu\gamma$  map stacked patches of  $1e+04$  brightest clusters,  $\mu$  value =  $5e-05$ , base sensitivity =  $0.20 \mu K \cdot \text{arcmin}$ ,  $r = 10p$



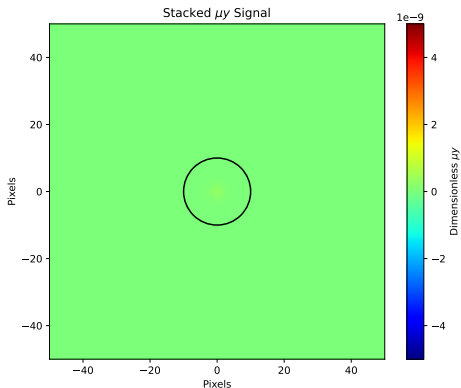
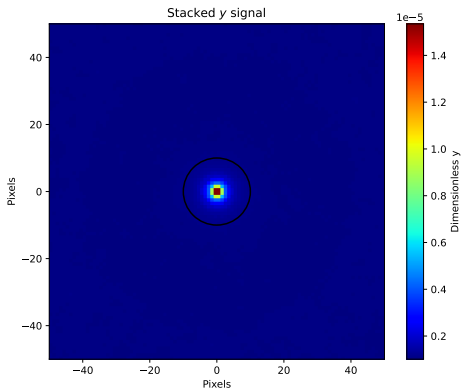
# Input stacked zoomed maps

4 layers: tSZ,  $\mu$ -distortion, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ )

$\mu = 2 \times 10^{-5}$ , freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$  pixels, stacked maps =  $10^4$

Input Data stacked patch of 10000 brightest clusters,  $\mu = 2\text{e-}05$ , base sensitivity =  $0.20 \mu K \cdot \text{arcmin}$ ,  $r = 10$



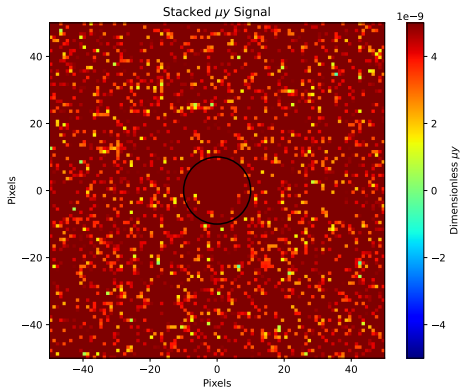
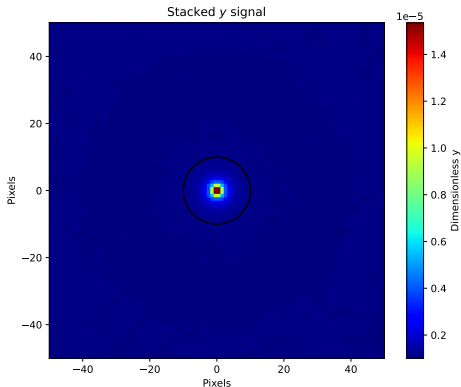
# ILC stacked zoomed maps

4 layers: tSZ,  $\mu$ -distortion, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ )

$\mu = 2 \times 10^{-5}$ , freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$  pixels, stacked maps =  $10^4$

P-ILC stacked patch of 10000 brightest clusters,  $\mu = 2\text{e-}05$ , base sensitivity =  $0.20 \mu K \cdot \text{arcmin}$ ,  $r = 10\text{p}$



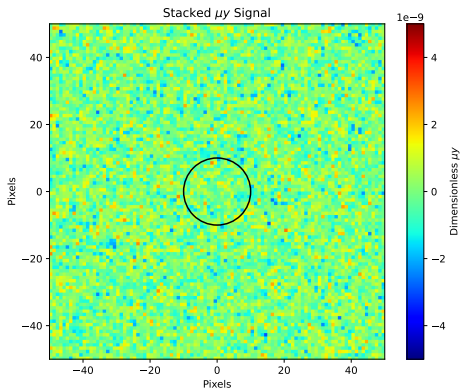
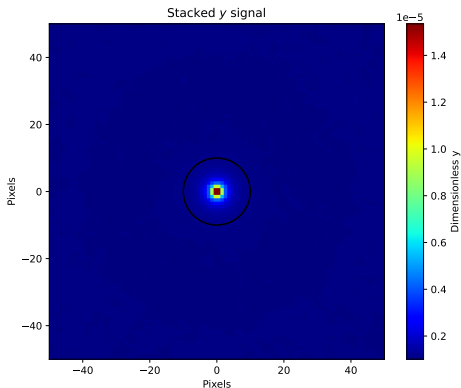
# CILC stacked zoomed maps

4 layers: tSZ,  $\mu$ -distortion, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ )

$\mu = 2 \times 10^{-5}$ , freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$  pixels, stacked maps =  $10^4$

P-CILC stacked patch of 10000 brightest clusters,  $\mu = 2\text{e-}05$ , base sensitivity =  $0.20 \mu K \cdot \text{arcmin}$ ,  $r = 10\text{p}$



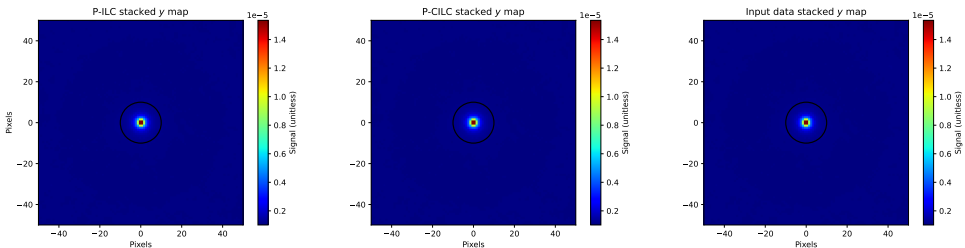
# Comparison $y$ stacked zoomed maps

4 layers: tSZ,  $\mu$ -distortion, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ )

$\mu = 2 \times 10^{-5}$ , freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$  pixels, stacked maps =  $10^4$

Comparison of  $y$  map stacked patches of  $1e+04$  brightest clusters,  $\mu$  value =  $2e-05$ , base sensitivity =  $0.20 \mu K \cdot \text{arcmin}$ ,  $r = 10p$



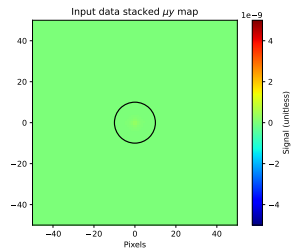
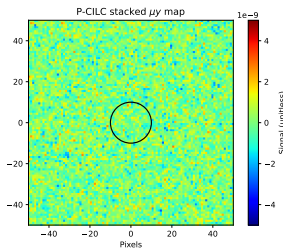
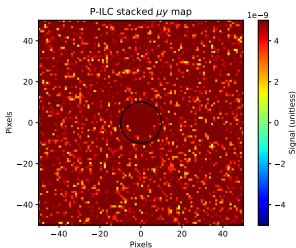
# Comparison $\mu\gamma$ stacked zoomed maps

4 layers: tSZ,  $\mu$ -distortion, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ )

$\mu = 2 \times 10^{-5}$ , freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$  pixels, stacked maps =  $10^4$

Comparison of  $\mu\gamma$  map stacked patches of  $1e+04$  brightest clusters,  $\mu$  value =  $2e-05$ , base sensitivity =  $0.20 \mu K \cdot \text{arcmin}$ ,  $r = 10p$



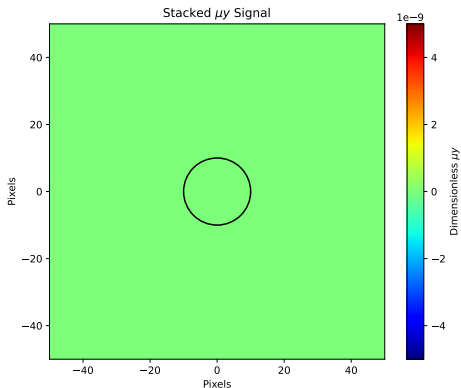
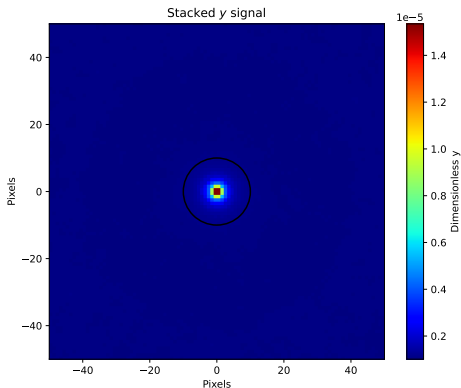
# Input stacked zoomed maps

3 layers: tSZ, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ )

Null test:  $\mu = 0$ , freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$  pixels, stacked maps =  $10^4$

Input Data stacked patch of 10000 brightest clusters,  $\mu = 0e+00$ , base sensitivity =  $0.20 \mu K \cdot \text{arcmin}$ ,  $r = 10p$



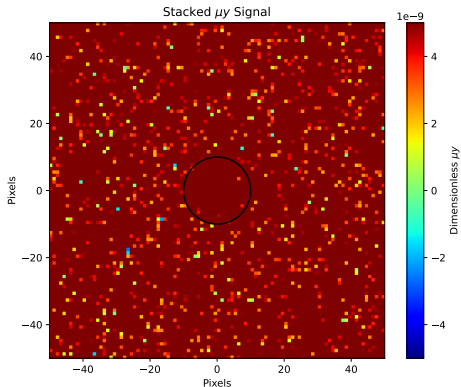
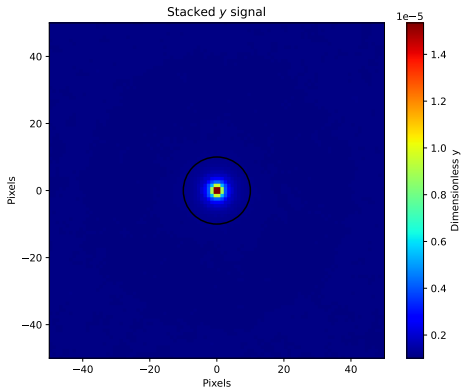
# ILC stacked zoomed maps

3 layers: tSZ, CMB and noise ( $0.2\mu\text{K}\cdot\text{arcmin}$ )

Null test:  $\mu = 0$ , freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$  pixels, stacked maps =  $10^4$

P-ILC stacked patch of 10000 brightest clusters,  $\mu = 0\text{e}+00$ , base sensitivity =  $0.20\mu\text{K}\cdot\text{arcmin}$ ,  $r = 10\text{p}$



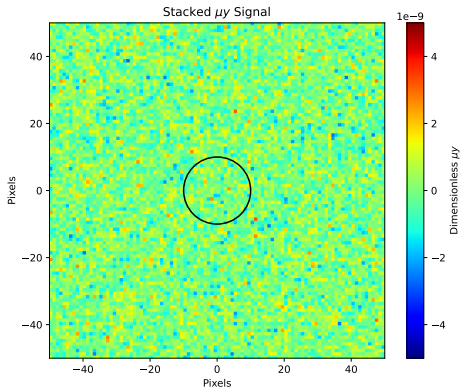
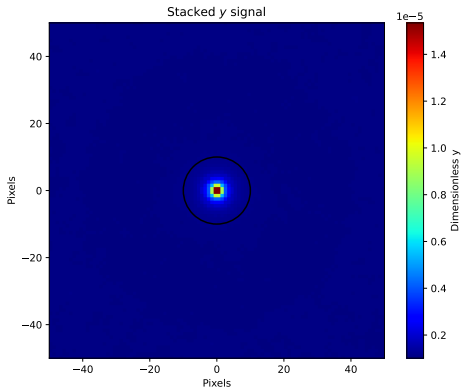
# CILC stacked zoomed maps

3 layers: tSZ, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ )

Null test:  $\mu = 0$ , freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$  pixels, stacked maps =  $10^4$

P-CILC stacked patch of 10000 brightest clusters,  $\mu = 0e+00$ , base sensitivity =  $0.20 \mu K \cdot \text{arcmin}$ ,  $r = 10p$



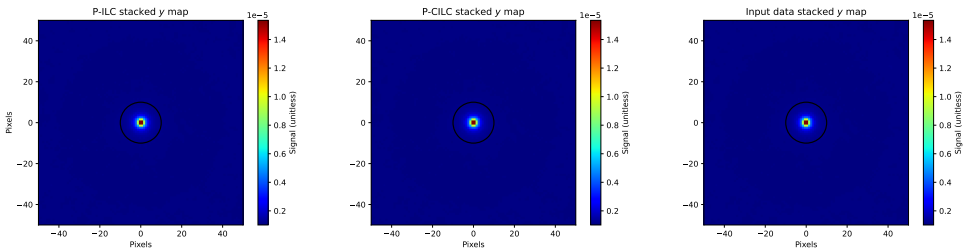
# Comparison $y$ stacked zoomed maps

3 layers: tSZ, CMB and noise ( $0.2\mu\text{K}\cdot\text{arcmin}$ )

Null test:  $\mu = 0$ , freqs = [30, 40, 90, 150, 220, 270] GHz

$r = 10$  pixels, stacked maps =  $10^4$

Comparison of  $y$  map stacked patches of  $1e+04$  brightest clusters,  $\mu$  value =  $0e+00$ , base sensitivity =  $0.20\mu\text{K}\cdot\text{arcmin}$ ,  $r = 10p$



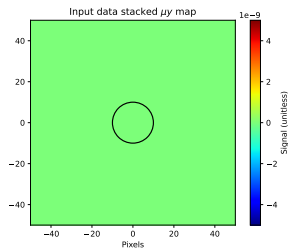
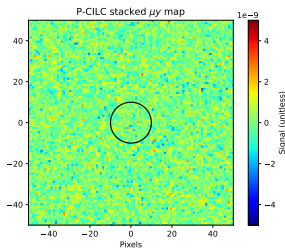
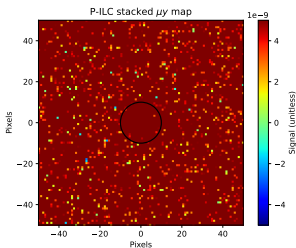
# Comparison $\mu\gamma$ stacked zoomed maps

3 layers: tSZ, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ )

Null test:  $\mu = 0$ , freqs = [30, 40, 90, 150, 220, 270] GHz

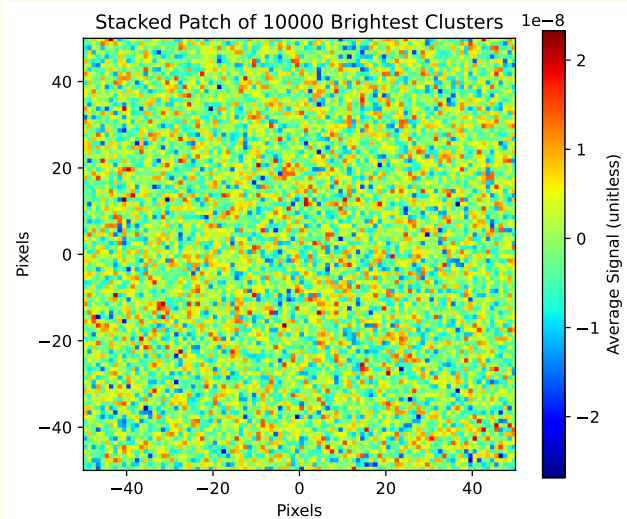
$r = 10$  pixels, stacked maps =  $10^4$

Comparison of  $\mu\gamma$  map stacked patches of  $1e+04$  brightest clusters,  $\mu$  value =  $0e+00$ , base sensitivity =  $0.20 \mu K \cdot \text{arcmin}$ ,  $r = 10p$



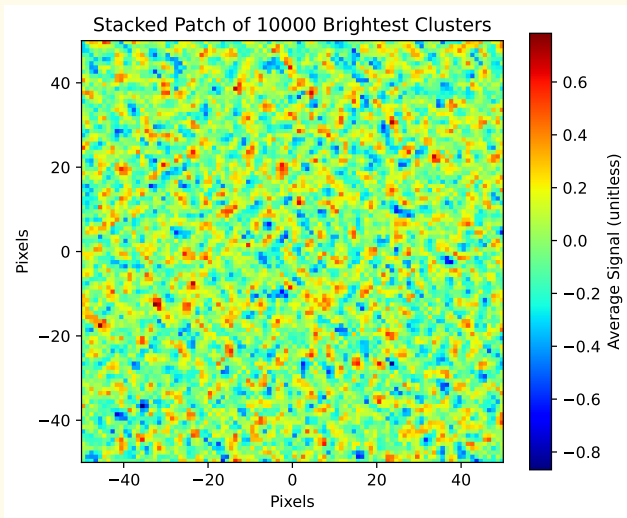
# P-ILC stacked zoomed maps (no $y$ input)

3 layers:  $\mu$ -distortion, CMB and noise ( $30\mu K \cdot \text{arcmin}$ )  
 $\mu = 2 \times 10^{-8}$ , freqs = [30, 40, 90, 150, 220, 270] GHz



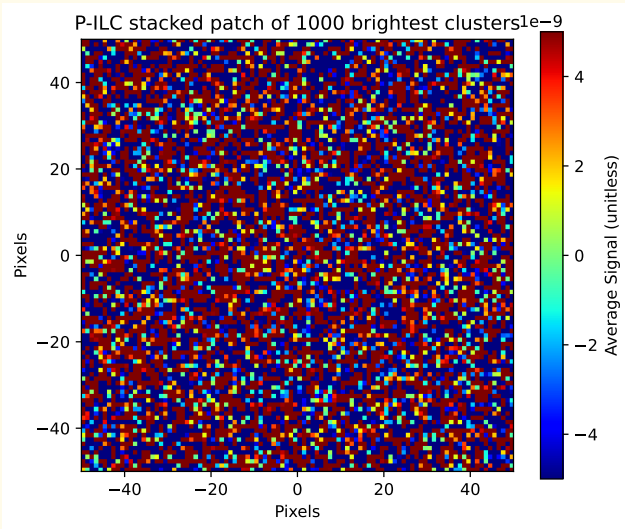
# H-ILC stacked zoomed maps (no $y$ input)

3 layers:  $\mu$ -distortion, CMB and noise ( $30\mu K \cdot \text{arcmin}$ )  
 $\mu = 2 \times 10^{-8}$ , freqs = [30, 40, 90, 150, 220, 270] GHz



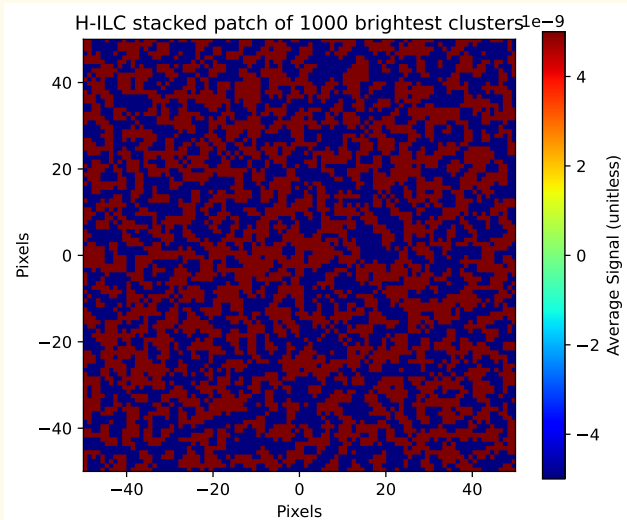
# P-ILC stacked zoomed maps (no $y$ input)

3 layers:  $\mu$ -distortion, CMB and noise ( $2\mu K \cdot \text{arcmin}$ )  
 $\mu = 2 \times 10^{-4}$ , freqs = [30, 40, 90, 150, 220, 270] GHz



# H-ILC stacked zoomed maps (no $y$ input)

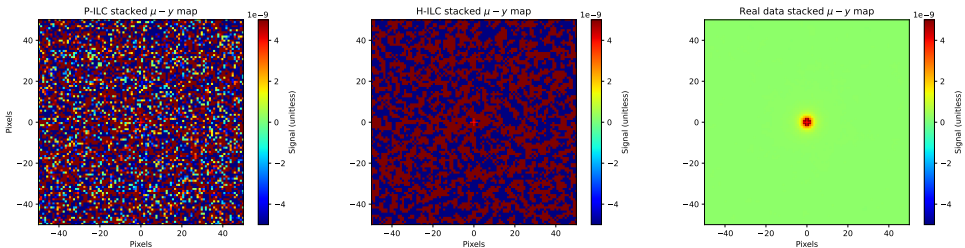
3 layers:  $\mu$ -distortion, CMB and noise ( $2\mu K \cdot \text{arcmin}$ )  
 $\mu = 2 \times 10^{-4}$ , freqs = [30, 40, 90, 150, 220, 270] GHz



# Comparison of stacked zoomed maps (no $y$ input)

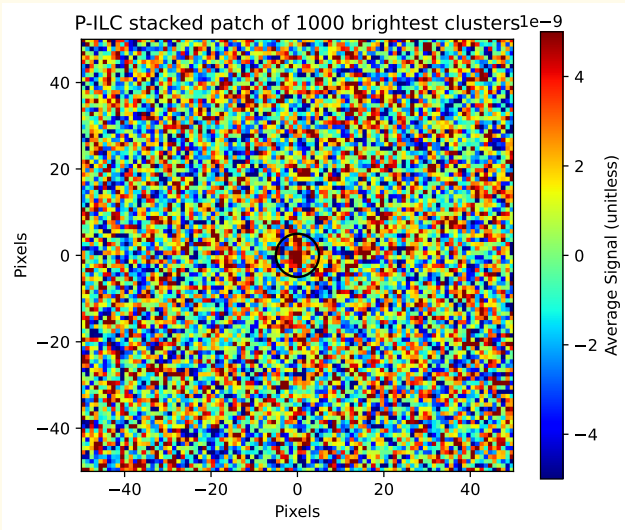
3 layers:  $\mu$ -distortion, CMB and noise ( $2\mu K \cdot \text{arcmin}$ )  
 $\mu = 2 \times 10^{-4}$ , freqs = [30, 40, 90, 150, 220, 270] GHz

Comparison of stacked patches of 1000 brightest clusters,  $\mu$  value =  $2e-04$



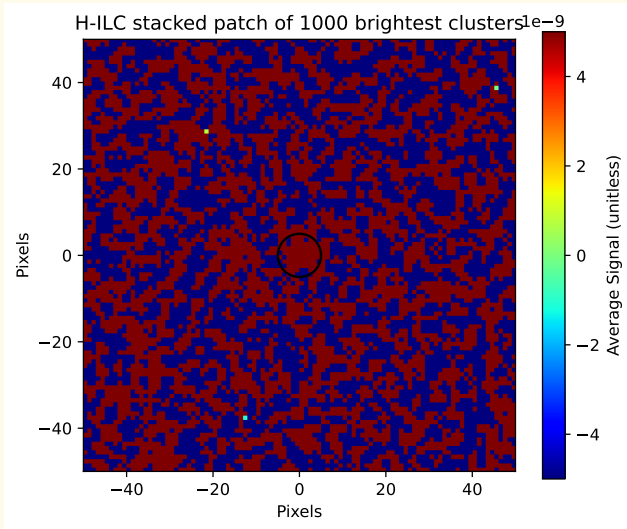
# P-ILC stacked zoomed maps (no $y$ input)

3 layers:  $\mu$ -distortion, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ )  
 $\mu = 2 \times 10^{-4}$ , freqs = [30, 40, 90, 150, 220, 270] GHz



# H-ILC stacked zoomed maps (no $y$ input)

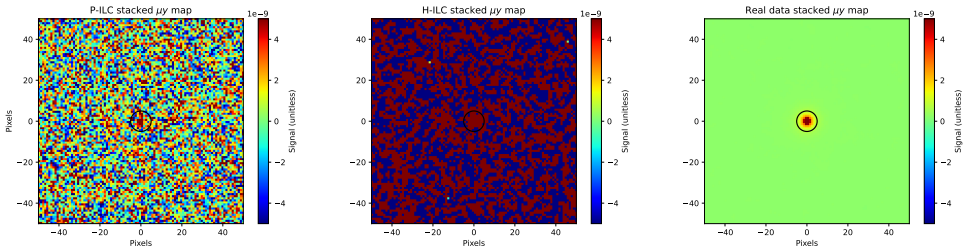
3 layers:  $\mu$ -distortion, CMB and noise ( $0.2\mu K \cdot \text{arcmin}$ )  
 $\mu = 2 \times 10^{-4}$ , freqs = [30, 40, 90, 150, 220, 270] GHz



# Comparison of stacked zoomed maps (no $y$ input)

3 layers:  $\mu$ -distortion, CMB and noise ( $0.2\mu\text{K}\cdot\text{arcmin}$ )  
 $\mu = 2 \times 10^{-4}$ , freqs = [30, 40, 90, 150, 220, 270] GHz

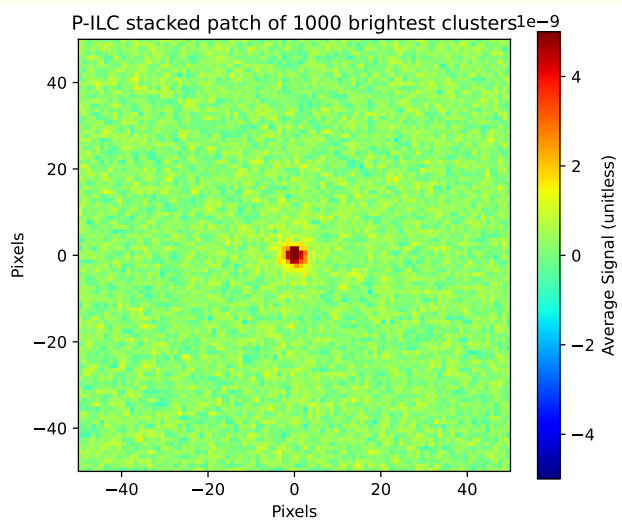
Comparison of stacked patches of 1000 brightest clusters,  $\mu$  value =  $2\text{e-}04$ , base sensitivity =  $0.20\mu\text{K}\cdot\text{arcmin}$



# P-ILC stacked zoomed maps (no $y$ input)

3 layers:  $\mu$ -distortion, CMB and noise ( $0.02\mu K \cdot \text{arcmin}$ )

$\mu = 2 \times 10^{-4}$ , freqs = [30, 40, 90, 150, 220, 270] GHz



# H-ILC stacked zoomed maps (no $y$ input)

3 layers:  $\mu$ -distortion, CMB and noise ( $0.02\mu K \cdot \text{arcmin}$ )

$\mu = 2 \times 10^{-4}$ , freqs = [30, 40, 90, 150, 220, 270] GHz

